

TABLE 2—EFFECT OF THE REFORM, SEPARATELY FOR EACH MIDDLE SCHOOL GPA QUINTILE

	School completion	University enrollment	University completion	School completion	University enrollment	University completion
	(1)	(2)	(3)	(4)	(5)	(6)
Bergen × Post	−0.018 (0.018)	−0.010 (0.020)	−0.014 (0.021)			
Bergen × Post × Quintile 1				−0.139 (0.039)	−0.118 (0.043)	−0.068 (0.047)
Bergen × Post × Quintile 2				−0.030 (0.037)	−0.006 (0.041)	0.019 (0.044)
Bergen × Post × Quintile 3				0.017 (0.037)	0.018 (0.041)	−0.044 (0.044)
Bergen × Post × Quintile 4				0.040 (0.038)	0.010 (0.041)	−0.013 (0.045)
Bergen × Post × Quintile 5				0.007 (0.036)	0.013 (0.039)	0.018 (0.042)
Observations	22,431	22,431	22,431	22,383	22,383	22,383

Notes: Estimates of parameters of a difference-in-difference regression of students’ outcome, where we compare the difference in outcomes between pre- and post-reform cohorts of academic high school students in Bergen to the same difference in other municipalities (Trondheim, Stavanger, Drammen, and Kristiansand) where no changes in the high school allocation system happened. *Bergen* is a dummy for attending high school in the Bergen area, *Post* is a dummy for starting high school after 2005, the year of the Bergen reform. The *Quintile* variables refer to quintiles of the middle school GPA distribution. The regression in columns 1–3 includes controls for municipality fixed effects, cohorts fixed effects, and municipality-specific linear trends in the month of birth. The regression in columns 4–6 includes additional controls for five dummies for being within each quintile of the middle school GPA distribution and their interaction with municipality fixed effect, cohort fixed effect, and municipality-specific linear trends in the month of birth.

(−1.8 percentage points), enrolling at university (−1 percentage points), and completing it (−1.4 percentage points). None of these estimated coefficients is precisely estimated, but this result tends to exclude any positive overall effect of the reform.

Because the reform differently affected individuals with different middle school GPAs, we also investigate whether we can recover some heterogeneity based on where students stand in the middle school GPA distribution of their cohort. We reestimate the difference-in-difference model by allowing for separate effects of the reform based on quintiles of middle school GPA distribution.³⁷ We report this heterogeneity analysis in columns 4–6 of Table 2. Students in the bottom quintile in Bergen seem to suffer some losses from being affected by the reform compared to students from the same cohorts in the same quintile in other municipalities. Individuals in the bottom quintile are those who are more likely to be excluded from competitive high school as a consequence of the reform (Section IVA). On the contrary, for high-grade students, the probability of attending more competitive high schools in Bergen is increasing. However, the effect of the reform for them is small and never statistically significant.

As our second method, we use the RDD framework to analyze the aggregate effects of the reform within Bergen. Figure 3 shows the proportion of students born in each week around January 1, 1989, completing high school (panel A), enrolling at university (panel B), and completing university (panel C) and a linear trend

³⁷ See equation (E.2) in Supplemental Appendix E for details.